

Project Title: From Roads to Classrooms: Educational Infrastructure Investment, Socioeconomic Equity, and Household Travel Behavior in Victoria (2023–2025)

Introduction

Victoria's 2023–24 State Budget directs significant funding toward infrastructure across education, transport, and community services, with school construction and upgrades making up the largest share of spatial investments [1]. At the same time, the Victorian Integrated Survey of Travel and Activity (VISTA) captures how households across Greater Melbourne and Geelong travel every day — for work, education, and beyond. Yet it remains unclear whether regions receiving more investment actually see different travel behaviors, whether a household's economic standing limits its transport options, and whether government infrastructure investment is being distributed across areas where Victorian households actually live. This project brings together three datasets — VISTA 2024–25, the State Budget Map Data, and the ABS SEIFA 2021 indexes — to explore these questions across Victoria's local government areas. The findings are relevant to policymakers, transport planners, and education authorities seeking to align infrastructure spending with community need.

Motivation

I am drawn to the question of whether government spending actually reaches the people who need it most. Victoria invests billions each year in regional infrastructure, but investment alone does not guarantee equity — if funding goes to already well-connected areas, or if economically disadvantaged households cannot access the transport options they need, the gap between intention and outcome widens. By combining travel survey data, budget investment records, and socioeconomic indexes, I want to explore whether infrastructure decisions across Victoria are spatially and socially aligned with real household travel demand, and what that means for equity in daily mobility.

Questions

1. How do daily travel patterns for work and education differ across regions with different levels of government infrastructure investment?
2. How does the economic status of a neighborhood shape the way its residents travel day to day?
3. How is government infrastructure investment distributed across Victorian areas relative to where households are located?

Data Source(s)

- **VISTA 2024–2025 (Relational):** Daily travel habits of Victorians. Includes temporal attributes (trip start times), categorical details (trip purpose, transport mode), and spatial attributes (origin/destination LGA). VISTA is the backbone dataset used to answer Q1, Q2, and Q3.
 - **Trips (19,864 rows × 65 cols):** https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/f8d93c74-cf02-48d7-a368-fc207c0b3c4f/download/trips_vista_2024_2025.csv
 - **Persons (6,404 rows × 53 cols):** https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/2a38732d-04dc-4af5-8a52-f281a8ceebb7/download/person_vista_2024_2025.csv
 - **Households (2,486 rows × 28 cols):** https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323/resource/1b73f806-1565-4243-96d7-78f9002c3e8f/download/household_vista_2024_2025.csv
- **State Budget Map Data 2023–24 (Spatial):** Contains geographical attributes (latitude and longitude) and investment values for 356 infrastructure projects across Victoria. Dominated by Education entries (193 rows) covering new school construction, upgrades, and early learning centres. Used alongside VISTA to answer Q1 (investment levels across regions) and Q3 (spatial distribution of investment relative to households).
 - **Budget Data (356 rows × 11 cols):** <https://discover.data.vic.gov.au/dataset/ed30724b-6ae7-4713-aff3-4f00b3666684/resource/71ca8b10-5a4e-483d-b3a3-a94421f4ef8b/download/state-budget-2023-24-budget-website-map-data.csv>
- **ABS SEIFA 2021 LGA Indexes (Tabular):** Provides ordinal attributes including socioeconomic scores and deciles (IRSD, IRSAD, IER, IEO) for all Australian LGAs. Primarily used to answer Q2 by defining the economic status of neighbourhoods. Also provides supporting socioeconomic context for Q1 and Q3.
 - **SEIFA LGA Indexes (xlsx):** <https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-areas-seifa-australia/2021/Local%20Government%20Area%2C%20Indexes%2C%20SEIFA%202021.xlsx>

References

- [1] Victorian Department of Treasury and Finance. (2023). *2023–24 Victorian State Budget – Infrastructure Investment Map*. Victorian Government. <https://discover.data.vic.gov.au/dataset/ed30724b-6ae7-4713-aff3-4f00b3666684> [Accessed: 15.03.2026].
- [2] Australian Bureau of Statistics. (2023). *Socio-Economic Indexes for Areas (SEIFA), Australia, 2021*. ABS. <https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-areas-seifa-australia/2021> [Accessed: 15.03.2026].
- [3] Department of Transport and Planning, Victoria. (2025). *Victorian Integrated Survey of Travel and Activity (VISTA) 2024–25*. Victorian Government Open Data. <https://opendata.transport.vic.gov.au/dataset/174fbf40-bef0-434a-8af0-891c7f3d2323> [Accessed: 15.03.2026].